

DIGITAL SIGNATURE SCHEME:-

A (digital) signature scheme consists of three probabilistic polynomial-time algorithms $(Gen, Sign, Verfy)$ such that:

1. Key-generation algorithm Gen

Input: Security parameter 1^n

Output: a pair of keys (pk, sk)

$Gen(1^n) \rightarrow (pk, sk)$

2. Signing algorithm $Sign$

Input: sk and message $m \in \{0, 1\}^*$

Output: signature σ

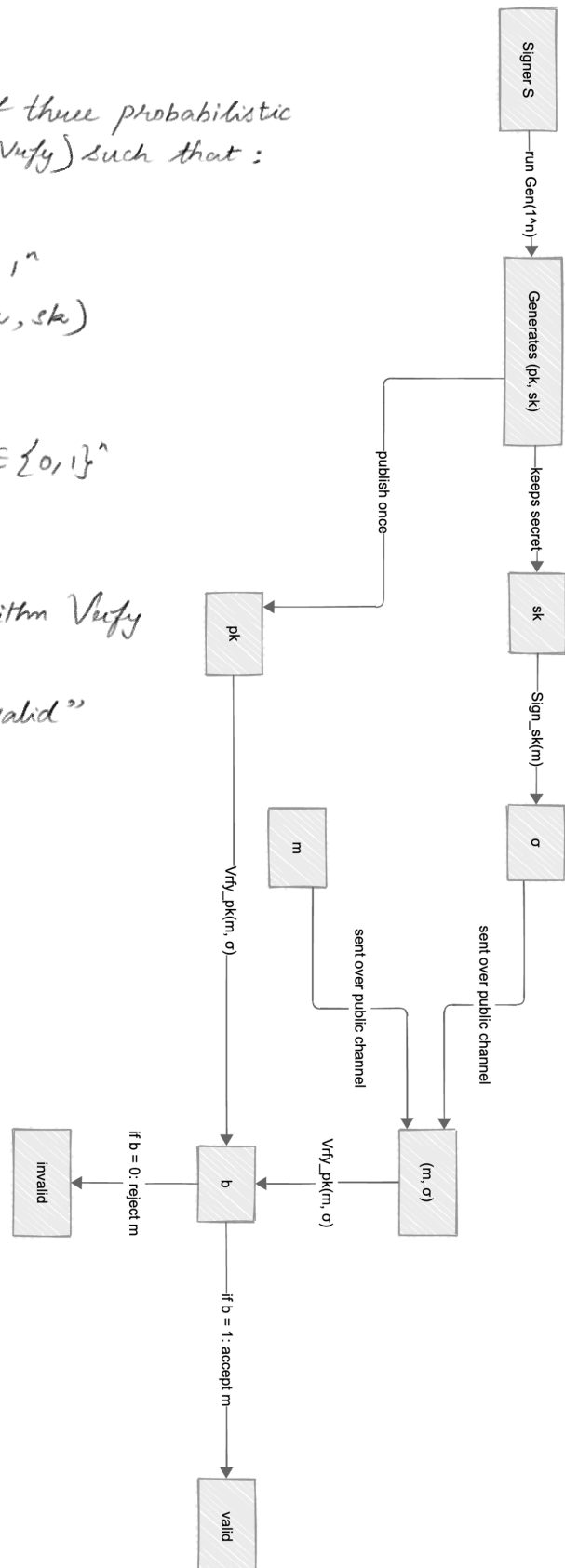
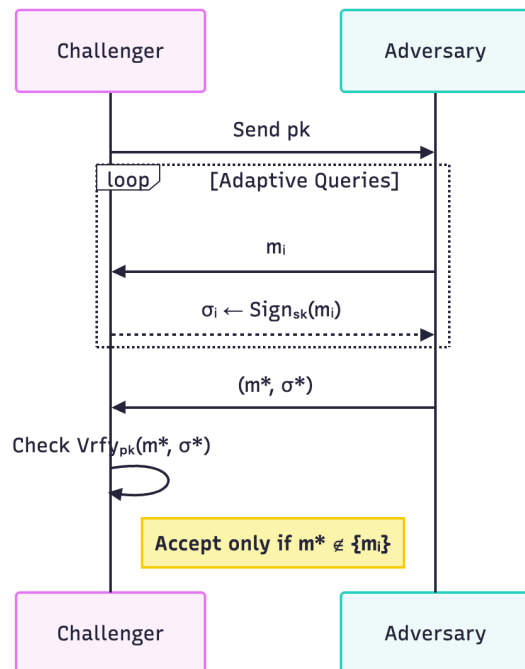
$Sign_{sk}(m) \rightarrow \sigma$

3. Deterministic verification algorithm $Verfy$

Input: pk, m, σ

Output: $1 = \text{"valid"}$; $0 = \text{"invalid"}$

$Verfy_{pk}(m, \sigma) \rightarrow b \in \{0, 1\}$

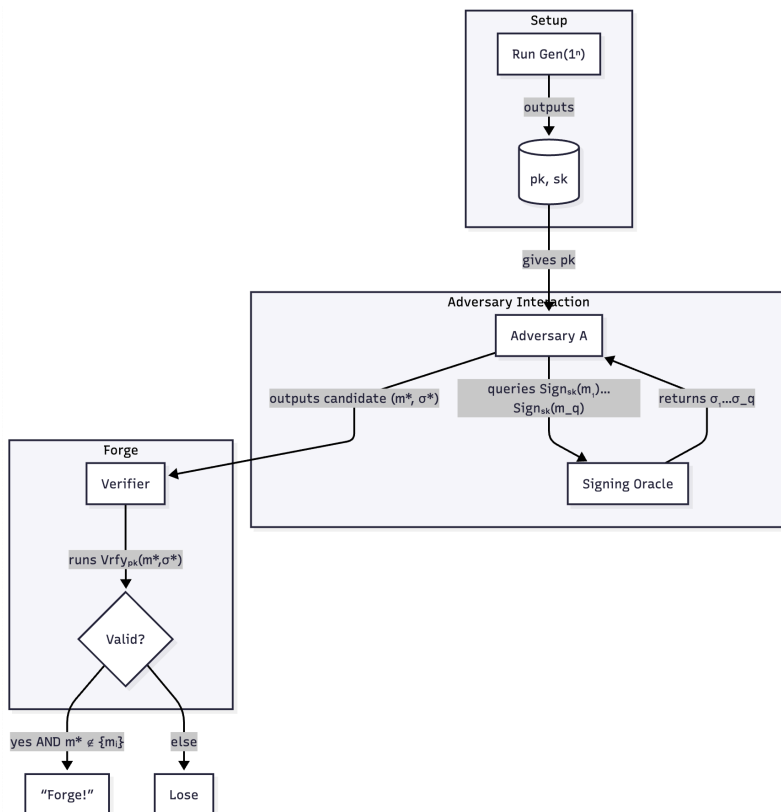


SECURITY OF SIGNATURE SCHEME :- Security of a signature scheme means that an adversary should be unable to output a forgery even if it obtain signatures on many other messages of its choice.

Let $\Pi = (\text{Gen}, \text{Sign}, \text{Verify})$ be a signature scheme and consider the following experiment for an adversary $\mathcal{A}$ and parameter n :

The signature experiment $\text{Sig-Forge}_{\mathcal{A}, \Pi}(n)$:

1. $\text{Gen}(1^n)$ is run to obtain (pk, sk)
2. Adversary $\mathcal{A}$ is given pk and access to an oracle $\text{Sign}_{sk}(\cdot)$. The adversary then outputs (m, σ) . Let Q denote the set of all queries that $\mathcal{A}$ asked its oracle.
3. $\mathcal{A}$ succeeds $\iff \text{Verify}_{pk}(m, \sigma) = 1$ and $m \notin Q$



A signature scheme $\Pi = (\text{Gen}, \text{Sign}, \text{Verify})$ is existentially unforgeable under an adaptive chosen message attack, or just secure, if for all probabilistic polynomial-time adversaries $\mathcal{A}$, there is a negligible function negl such that:

$$\Pr[\text{Sig-Forge}_{\mathcal{A}, \Pi}(n) = 1] \leq \text{negl}(n)$$

Event that $\mathcal{A}$ successfully outputs a valid signature σ^* on some message m^*

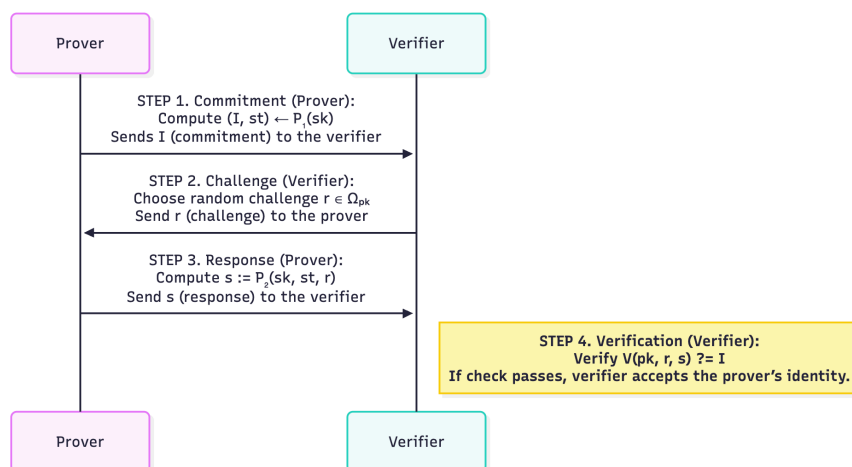
$\mathcal{A}$ can ask for signatures on messages of its choice, based on previously received signatures

IDENTIFICATION SCHEME:- An identification scheme is an interactive protocol used to prove the identity of a party (the prover) to another party (the verifier) securely.

The process ensures the verifier is convinced they're interacting with intended party, not an impostor.

USED IN PUBLIC KEY SETTING:-

- The verifier only knows the prover's public key.
- The prover holds a private key. no secret information is shared in advance.
- This setting is crucial for modern authentication where shared secrets (like passwords) are not viable or secure.



THREE ROUND IDENTIFICATION PROTOCOL

Key elements:

- P, P_a : prover's algorithms
- V : Verifier's check
- Ω_{pk} : challenge space, depends on the public key
- sk : prover's private key
- pk : prover's public key
- st : state maintained by prover
- I : Initial message (commitment)
- r : random challenge
- s : prover's response

Commit Phase:

The prover picks a random value and creates a commitment I , together with a state st (possibly including the random choice). The commitment I is sent to the verifier.

Challenge Phase:

The verifier chooses a random challenge r from a set determined by pk . The challenge is sent to the prover.

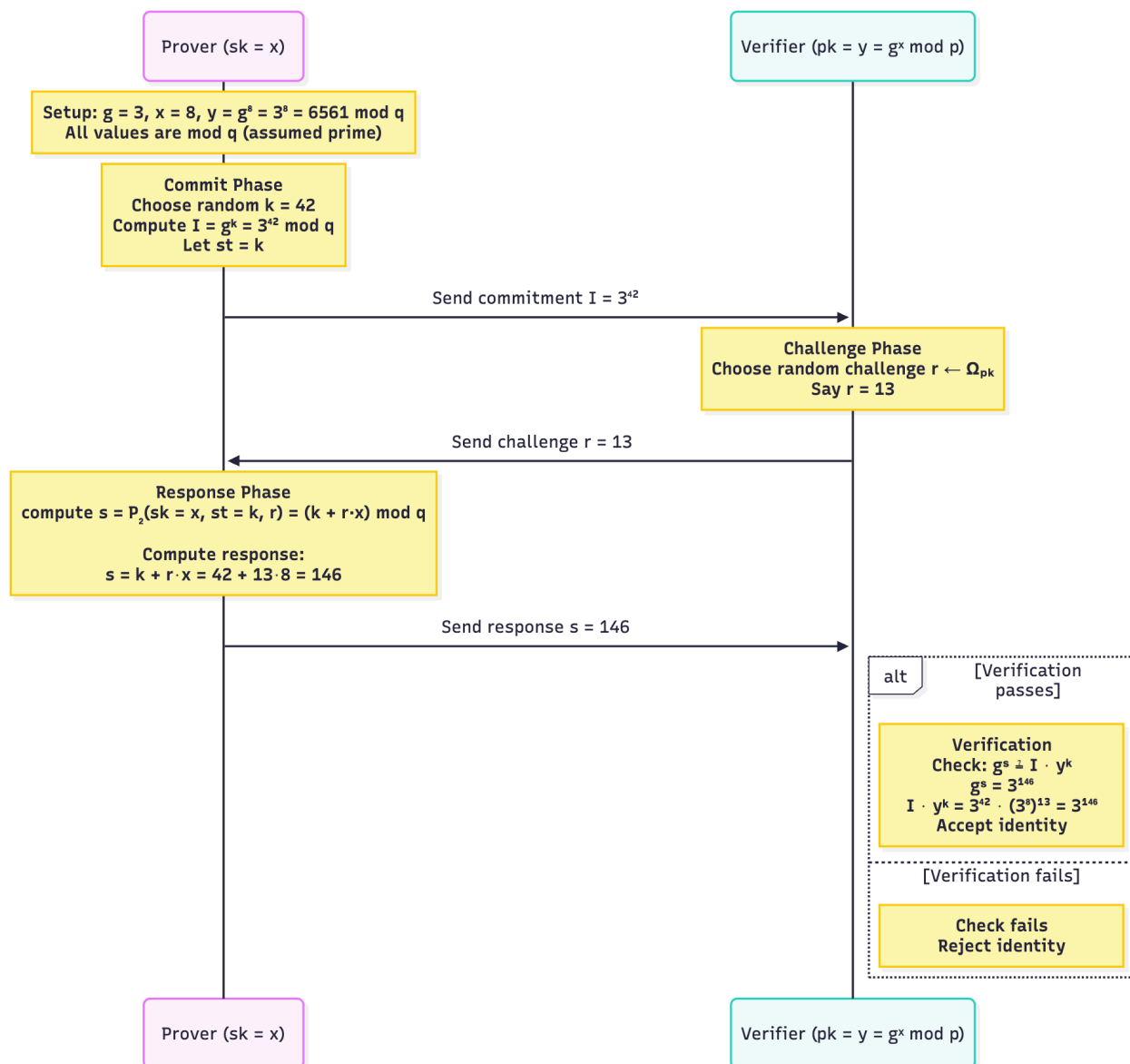
Response Phase:

Using their private key, the state st , and the challenge r , the prover computes a response s and sends it to the verifier.

Verification:

The verifier checks if $V(pk, r, s) = I$. If so, accept; otherwise, reject.

EXAMPLE



Security Requirements

1. Correctness:

A legitimate prover who follows the protocol with the correct private key and steps will always be accepted by the verifier.

2. Nondegeneracy:

- There must be many possible initial messages I .
- For any fixed secret key sk and initial message I , the chance $P_1(sk)$ outputs I should be negligible. $\forall sk \text{ and fixed } I, \Pr_{(I', sk) \leftarrow P_1(sk)} [I' = I] \leq \text{negl}(n)$
- Ensures that no I is likely to occur, preventing replay attacks.

3. Security Against Eavesdroppers:

- Even if an attacker passively observes many protocol runs (sees many triples (I, r, s)), they cannot successfully impersonate the prover.

Attack model: Eavesdropping Oracle.

- The adversary can query an oracle Trans_{sk} that gives them complete transcripts (I, r, s) from honest runs of the protocol.
- The adversary sees as many examples as desired.
- Later, the adversary tries to produce an initial message I , gets a challenge r , and must provide s so $V(pk, r, s) = I$

Goal: Without the secret key, it should be infeasible for any attacker to succeed at this challenge, even after many oracle queries

Let $\Pi = (\text{Gen}, P_1, P_2, V)$ be an identification scheme, and consider the following experiment for an adversary $\mathcal{A}$ and parameter n :

The identification experiment $\text{Iden}_{\mathcal{A}, \Pi}(n)$:

1. $\text{Gen}(1^n)$ is run to obtain keys (pk, sk) .
2. Adversary $\mathcal{A}$ is given pk and access to an oracle Trans_{sk} that it can query as often as it likes.
3. At any point during the experiment, $\mathcal{A}$ outputs a message I . A uniform challenge $r \in \Omega_{pk}$ is chosen and given to $\mathcal{A}$, who responds with some s . ($\mathcal{A}$ may continue to query Trans_{sk} even after receiving r .)
4. The experiment outputs 1 if and only if $V(pk, r, s) \stackrel{?}{=} I$.

An identification scheme $\Pi = (\text{Gen}, P_1, P_2, V)$ is secure against a passive attack (or) just secure, if for all probabilistic polynomial time adversaries $\mathcal{A}$, there exist a negligible function negl such that:

$$\Pr[\text{Iden}_{\mathcal{A}, \Pi}(n) = 1] \leq \text{negl}(n)$$

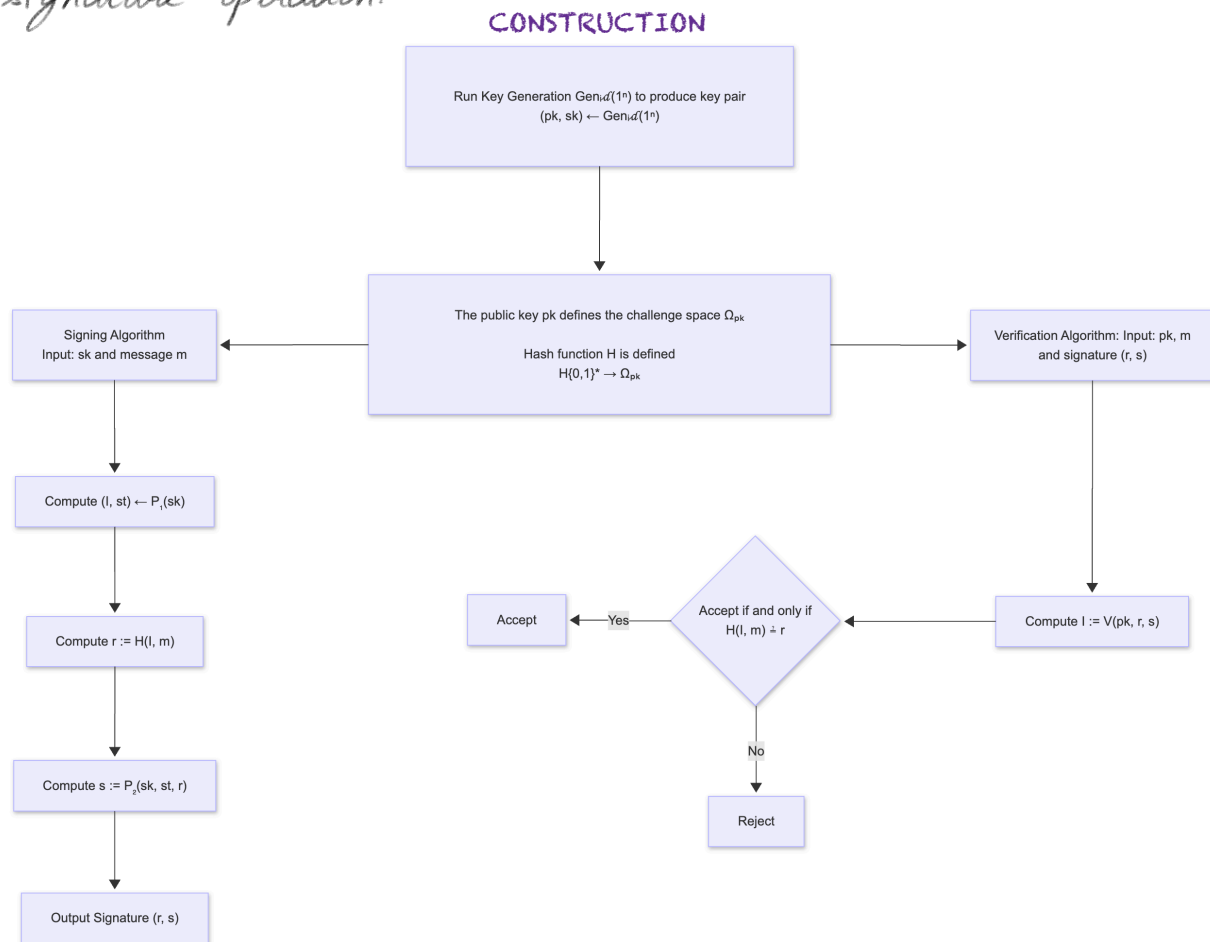
Passive security covers attacks where the adversary only observes honest interactions, not when it actively interferes.

Active security (not needed here) would also consider adversaries impersonating the verifier and sending malicious values.

FROM IDENTIFICATION SCHEMES TO DIGITAL SIGNATURES: FIAT-SHAMIR TRANSFORM

The Fiat-Shamir transform converts an (interactive) identification scheme into a (non-interactive) digital signature scheme.

The trick is to let the signer simulate the verifier using a hash function, turning the interactive steps into a single signature operation.



A signature (r, s) is “bound” to a specific message m because r is a function of both l and m ; changing m thus results in a completely different r . If H is modeled as a random oracle mapping inputs uniformly onto Ω_{pk} , then the challenge r is uniform; intuitively, it will be just as difficult for an adversary (who does not know sk) to find a valid signature (r, s) on a message m as it would be to impersonate the prover in an honest execution of the protocol.

Let $\Pi = (\text{Gen}, P_1, P_2, V)$ be a secure identification scheme, and let $\Pi' = (\text{Gen}', \text{Sign}, \text{Verify})$ be the Fiat-Shamir signature scheme obtained from Π by applying the Fiat-Shamir transform to it.

If Π is secure and $H: \{0,1\}^* \rightarrow \mathbb{Z}_{p_k}$ is modelled as a random oracle, then Π' is secure.

To prove: $\forall$ PPT adversary A' against Π' , $\exists$ a negligible function $\text{negl}(n)$ such that

$$\Pr[\text{Sig-forg}_{A', \Pi'}(n) = 1] \leq \text{negl}(n)$$

Goal: We show that if any adversary A' can forge a signature in Π' , then we can build an adversary that breaks the underlying identification scheme. Which is a contradiction (since Π is secure). $\therefore$ Forging a signature must be infeasible.

PROOF:

Let $q = q(n)$ be the number of queries A' makes to the random oracle H .

Assume A' queries each (I_i, m_i) only once.

If (u, s) is a forgery on m , then A' must have queried $H(I, m)$ where $I = V(p_k, u, s)$.

We construct an efficient adversary A that uses A' as a subroutine and attacks the identification scheme Π :

A is given pk and oracle access to honest protocol transcripts Trans_{sk} .

A will simulate the behavior of A' as if operating in the real signature scheme.

- (i) Pick $j \in \{1, 2, \dots, q\}$ uniformly at random.
- (ii) When A' queries $H(I_i, m_i)$
 - if $i = j$
 - A sends I_j (commitment) to the ID game
 - Receives a real challenge e_j from the verifier
 - Sets $H(I_j, m_j) = e_j$ as response to A'
 - if $i \neq j$
 - A picks uniformly random e_i from Σ_{pk}
 - Sets $H(I_i, m_i) = e_i$
- iii) If A' requests a signature on m
 - A queries Trans_{sk} to get an honest transcript (I, e, s)
 - Outputs (e, s) as the signature.
- iv) Eventually A' outputs a forgery (m^*, e^*, s^*) such that $\text{Verify}(pk, m^*, e^*, s^*) = 1$.
So $I^* = V(pk, e^*, s^*)$ and $H(I^*, m^*) = e^*$

If $(I^*, m^*) = (I_j, m_j)$, A uses s^* as the forged response to the challenge x_j for I_j in the ID experiment.

If not, A aborts.

Let $p = \Pr[\text{Sig-forg}_{A, \Pi'}(n) = 1] \rightarrow$ adversary chance to forge.

The probability that (I^*, m^*) matches (I_j, m_j) is exactly $1/q$ (since j is chosen uniformly at random).

So, the probability that A breaks the ID scheme is at least

$$\Pr[\text{Ident}_{A, \Pi}(n) = 1] \geq \frac{1}{q(n)} \cdot p - \text{negl}(n)$$

$$\Rightarrow \Pr[\text{Sig-forg}_{A, \Pi'}(n) = 1] \leq q(n) \cdot \Pr[\text{Ident}_{A, \Pi}(n) = 1] + \text{negl}(n)$$

If Π is secure, $\Pr[\text{Ident}_{A, \Pi}(n) = 1]$ is negligible.

$$\Rightarrow \Pr[\text{Sig-forg}_{A, \Pi'}(n) = 1] \leq q(n) \cdot \text{negl}(n) + \text{negl}(n)$$

Since $q(n)$ is polynomial, p is also negligible

$$\Rightarrow \Pr[\text{Sig-forg}_{A, \Pi'}(n) = 1] \leq \text{negl}(n)$$

$\therefore \Pi'$ is secure.

Schnorr Identification Scheme

Schnorr Identification Scheme Relies on the **discrete logarithm problem (DLP)**: given a cyclic group G of order q , a generator g , and an element $gy \in G$, it's hard to compute x such that $y = g^x$

A probabilistic algorithm $G(1^n)$ outputs:

- A cyclic group G
- A prime order q (with $|G| = n$)
- A generator g of G

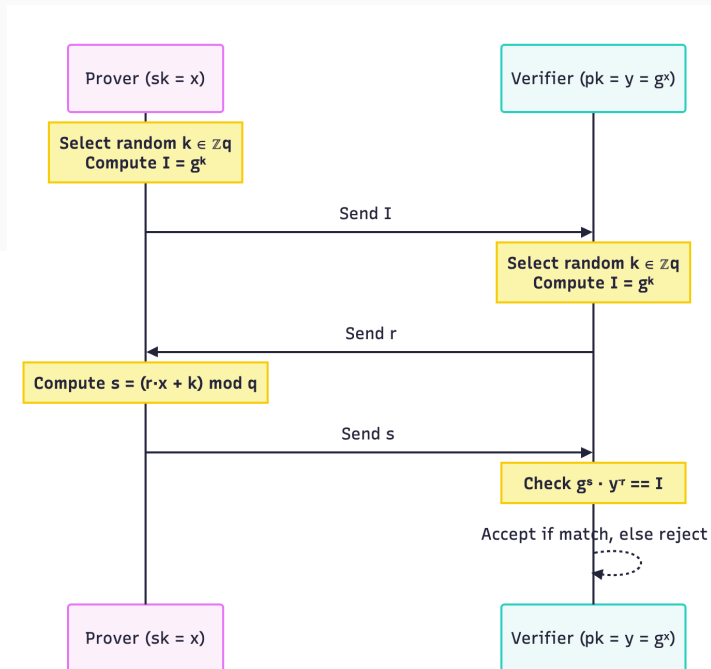
Prover picks secret x randomly from $\mathbb{Z}_q$

Sets public key $y = g^x$

Correctness holds because

$$g^s \cdot y^{-r} = g^{sx+k} \cdot (g^x)^{-r} \\ = g^{sx+k-ux} = g^k = I$$

Thus if both parties are honest, the identity is always confirmed.



Non-Degeneracy

The commitment $I = g^k$ is **uniformly random** in G for random k , so the protocol doesn't leak information and can't be predicted (which is a key requirement for security).

Simulatability (Passive Security)

Key Insight

- **Passive eavesdropping** is useless against Schnorr: anyone can simulate valid protocol transcripts without knowing the private key!
- **How?** The adversary does not need to eavesdrop; they can:
 1. Pick r, s independently and uniformly at random from $\mathbb{Z}_q$
 2. Set $I = g^s \cdot y^{-r}$
 3. The tuple (I, r, s) looks like a real protocol run

Reason:

- Honest and simulated transcripts have the same distribution: commitment I is uniform in G , r and s are independent and uniform in $\mathbb{Z}_q$.

Active Security, Special Soundness & Discrete Logarithm

Why Double Challenge Reveals the Secret

If an adversary can successfully respond to two different challenges for the same commitment I , it breaks the scheme:

- Suppose for r_1, r_2 :

$$g^{s_1} \cdot y^{-r_1} = I = g^{s_2} \cdot y^{-r_2} \Rightarrow g^{s_1 - s_2} = y^{r_1 - r_2}$$

- Take discrete logs base g :

$$\begin{aligned} s_1 - s_2 &\equiv x(r_1 - r_2) \pmod{q} \\ \Rightarrow x &\equiv (s_1 - s_2) \cdot (r_1 - r_2)^{-1} \pmod{q} \end{aligned}$$

- **Interpretation:** Two different valid responses to two challenges yield the secret x , contradicting DLP's assumed hardness.

THEOREM: If the discrete logarithm problem is hard relative to $\mathcal{G}$, then the Schnorr identification scheme is secure.

i.e., for any probabilistic polynomial time (PPT) adversary $\mathcal{A}$,

$$\Pr[\text{Ident}_{\mathcal{A}, \Pi}(n) = 1] \leq \text{negl}(n)$$

PROOF: Let G be a cyclic group of prime order q and generator g . Let $x \in \mathbb{Z}_q$, $y = g^x$ be the secret and public key respectively.

Suppose we have a probabilistic polynomial-time adversary A that attacks the Schnorr identification ^(PPT) scheme Π .

We build another PPT algorithm A' that solves the discrete logarithm problem in G .

The construction of A' uses A as a subroutine is as follows:

First A' runs $A(pk)$, simulating responses to all queries A makes to the transcript oracle Trans_{sk} exactly as in the honest protocol.

i.e., A' produces transcripts (I, r, z) where $I = g^k$, $r \in \mathbb{Z}_q$ is chosen uniformly at random and $z = rx + k \bmod q$.

This simulation is perfect and indistinguishable from real executions because A' can generate any number of such transcripts without knowing x .

During the first run, A eventually outputs an initial message I . Algorithm A' then selects a uniform random challenge $r_1 \in \mathbb{Z}_q$, sends r_1 to A and receives the response z_1 .

Next A' restarts $A(pk)$ using the same randomness w as the first run except it chooses a fresh

independent challenge $u_2 \in \mathbb{Z}_q$. $\mathcal{A}$ outputs the same initial message I and returns response s_2 to the challenge u_2 .

If the responses satisfy the verification equations, then we have

$$g^{s_1} \cdot y^{-u_1} = I \text{ and } g^{s_2} \cdot y^{-u_2} = I$$

and if $u_1 \neq u_2$, then $\mathcal{A}'$ computes the discrete logarithm x of y as $x = (s_1 - s_2)(u_1 - u_2)^{-1} \bmod q$ else output failure.

$$\begin{aligned} g^{s_1} \cdot y^{-u_1} &= g^{s_2} \cdot y^{-u_2} \\ \Rightarrow g^{s_1} \cdot g^{-xu_1} &= g^{s_2} \cdot g^{-xu_2} \\ \Rightarrow g^{s_1 - xu_1} &= g^{s_2 - xu_2} \\ \Rightarrow s_1 - xu_1 &= s_2 - xu_2 \\ \Rightarrow s_1 - s_2 &= x(u_1 - u_2) \\ \Rightarrow x &= (s_1 - s_2)(u_1 - u_2)^{-1} \end{aligned}$$

Let

$$V(w, u) = \begin{cases} 1 & \text{if } \mathcal{A} \text{ responds correctly to } u \text{ under randomness } w \\ 0 & \text{otherwise.} \end{cases}$$

For a fixed w , let $\delta_w := \Pr_{u \in \mathbb{Z}_q} [V(w, u) = 1]$ be the probability over random challenge u that $\mathcal{A}$ answers correctly against randomness w .

Then the advantage that $\mathcal{A}$ has in impersonation

$$\text{is } \delta_n := \Pr_{w, u} [V(w, u) = 1] = \mathbb{E}_w [\delta_w] = \Pr[\text{Ident}_{A, \Pi}(n) = 1]$$

The algorithm $\mathcal{A}'$ succeeds if it obtains two correct responses s_1, s_2 for two distinct challenges $u_1 \neq u_2$.

Thus,

$$P_u [D\text{Log}_{A', G}(n) = 1] = P_{u, u_1, u_2} [V(w, u_1) \wedge V(w, u_2) \wedge u_1 \neq u_2]$$

$$\therefore P_{u, u_1, u_2} [V(w, u_1) \wedge V(w, u_2) \wedge u_1 \neq u_2] \geq P_{u, u_1, u_2} [V(w, u_1) \wedge V(w, u_2)] - P_u [u_1 = u_2]$$

$\therefore u_1$ and u_2 are uniform and independent over $\mathbb{Z}_q$,

$$\text{we have } P_u [u_1 = u_2] = 1/q$$

Also by independence,

$$P_{u, u_1, u_2} [V(w, u_1) \wedge V(w, u_2)] = \mathbb{E}_w [\delta_w^2] \text{ which is}$$

the expected squared probability of success over fixed randomness w .

By Jensen's inequality, which states that for any convex function ϕ , $\mathbb{E}[\phi(x)] \geq \phi(\mathbb{E}[x])$ and since x^2 is convex, we have

$$\mathbb{E}_w [\delta_w^2] \geq (\mathbb{E}_w [\delta_w])^2 = \delta(n)^2$$

$$\therefore P_u [D\text{Log}_{A', G}(n) = 1] \geq \delta(n)^2 - 1/q$$

If the discrete logarithm problem is hard relative to G then $\Pr[\text{Dlog}_{A', n}(n) = 1]$ is negligible

Since $1/q$ is negligible because the size of q grows exponentially with n .

$\Rightarrow \delta(n^2)$ is negligible and hence $\delta(n)$ itself is negligible, so Π is a secure identification scheme.

Hence no PPT adversary impersonates with non-negligible probability, proving security of Schnorr ID.